Q. *If a fixed effect is taken to be not statistically significant should it be dropped from the model?*
My short answer is that if it is a main effect then, certainly **NO**, it should not be dropped from the model, even if it is a nuisance factor. I will consider interactions later.
1. What does statistically non-significant mean? It means that the data cannot exclude the possibility that the null hypothesis (H0) is true, which is typically that all the effects are 0, given the Type 1 error we have chosen. My view on a classical hypothesis testing is that it is a statement of whether or not there is evidence to accept that the effects not 0, with our Type 1 error giving an indication of the strength of evidence required to accept this[1].
2. ***It is also the case***, that a wide range of possible non-zero values for the effects would be compatible with the data; if the power of the study is weak for these effects, then some of these possible values may be large! So, there are a large number of possible H0 that would also be non-significant other than all the effects are 0.
3. If I remove a fixed effect from the model, then all the other parameters that I estimate are ***conditional upon*** that effect being truly 0. If I include it in the model, then I have made the best possible attempt to cater for these effects, whatever they may be, and as far as my data will allow. Therefore, I would prefer to take the 'high ground' by not unnecessarily assuming something is 0, when I cannot be certain that they are 0.
4. In practice, if another paper (perhaps more powerful) shows effects that are non-zero, how am I to defend unnecessarily making an assumption that they do not exist, and what value will be placed on my estimates? Will a reader trust my conclusions?
5. I would always include main effects, as spending a few d.f. on reaching the 'high ground' is worth it. If the power of the data cannot afford them then the design is very poor, and the analysis already has problems. Sometimes interactions can take up large numbers of d.f. which may be undesirable, and also, at what level of interactions (i.e. 2-way, 3-way) does one stop if one has several factors? It is always important to explore data, but often there are ways of addressing some of these problems and some ways are introduced in Lecture 5.
6. The typical default H0 of all effects are 0 is used because it is generic, readily expressed and tested, and does have a qualitative interpretation of whether a possible phenomenon is real or not. A personal world view is that many things affect traits, some of them only trivially and would only be uncovered with huge data. I have read that a typical trait with no heritability is the number of scars one acquires in life, but while it may be small, will it be 0 - if one considers that there is genetic variation in behavioural traits?

[1] If I have no evidence, I don't want to waste time on reporting such results UNLESS it was an important objective of the study, in which case the estimate and its standard error are always valuable!